Ivan Popovych

Software Developer– Machine Learning & NLP

Contact Information

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Professional Summary

Motivated Computer Science student with hands-on experience in machine learning, NLP, and full-stack web development. Skilled in building and fine-tuning ML models, utilizing deep learning frameworks like TensorFlow and Hugging Face Transformers. Experienced in integrating APIs, creating responsive UIs with React, and deploying scalable backends. Known for strong problem-solving, adaptability, and teamwork.

Education

- **Uzhhorod National University I Computer Science B.S.** *Uzhhorod, Ukraine / Online*
  Sep 2022 – Jun 2026
- **Hochschule Darmstadt (h_da) I Exchange** *Darmstadt, Germany*
  Oct 2023 – Feb 2025

Experience

- **Student Assistant – ML / NLP** *Hochschule Darmstadt* I Darmstadt, Germany I **Dec 2024 - Present**
  - Development of machine learning models for **text classification**.
  - Leveraged different NLP libraries using technologies such as **SpaCy**, **NLTK, Tensorflow, Keras**, **Hugging Face Transformers**, **TextBlob** for feature extraction and preprocessing.
  - Collaborated with **academic staff** to improve existing systems and datasets, processing **100,000+** text samples for model evaluation.
- **Fullstack Developer** *Freelance* I Remote I **Jul 2024 - Present**
  - Designed and developed responsive web applications with **React**, **TypeScript**, **MaterialUI** and **Tailwind**, ensuring optimal UX across devices.
  - Integrated multiple third-party APIs for dynamic data retrieval.
  - Built a scalable backend with **Node.js**, **Express.js**, **MongoDB** and **REST API**, improving DB performance.
- **Intern - IT Assistant** *Uzhhorod City Council* I Uzhhorod, Ukraine I **Jan – Apr 2022**
  - Collected and organized data for a future municipal database, analyzing information from **10+ organizations**.
  - **Managed** and optimized content for **3+** social media platforms, increasing the number of followers by **30%.**
  - Supported the development of **digital tools**, streamlining administrative workflows in a government setting.

Skills

- **Python:** Flask, Numpy, Pandas, MatplotLib
- **Machine Learning:** ScikitLearn, Tensorflow, CNNs, Deep Learning, Computer Vision, Feature Engineering
- **NLP:** SpaCy, NLTK, Hugging Face, TextBlob
- **Frontend:** HTML, CSS, JavaScript, TypeScript, React, MaterialUI, Tailwind
- **Backend:** Node.js, Express.js, REST API
- **Databases:** MongoDB, SQL, SPARQL
- **Tools:** Docker, Git, GitLab CI/CD
- **Soft Skills:** Quick learning, Problem Solving, Teamwork, Adaptability, Flexibility, Innovation

Academic Projects

- **Brain Computer Interface ML Application** *Hochschule Darmstadt*
    - Developed an ML model to detect eye blinks with **~90%** accuracy using TensorFlow.
    - Analyzed and preprocessed brainwave datasets consisting of **30,000+** samples for model training.
    - Implemented a model that uses real time electrical brain impulses from a Brain Computer Interface.
- **AI Movie Recommender** *Hochschule Darmstadt*
    - Built a Flask-based web app that recommends movies using **SPARQL queries** and **NLP** techniques.
    - Enhanced user experience by analyzing text inputs, increasing recommendation accuracy by 15%.
    - Optimized data retrieval, reducing response time by **80%** for better user experience.
- **IForms | Web App with sentiment analysis** *Hochschule Darmstadt*
    - Built a platform for creating and analyzing questionnaires using React, TS, MaterialUI (Frontend) and Express.js, Node.js, Natural (Backend).
    - Integrated sentiment analysis, improving actionable feedback accuracy by **20%**.
- **Dashly | Multi User Widgets Web App** *Hochschule Darmstadt*
    - Designed and implemented the backend for a multi-user web application for customizable dashboards with widgets (Weather, Calendar, RSS, GitHub Issues, etc.).
    - Developed a multi-user authentication system, reducing login time by **90%** and improving overall user engagement.
    - Technologies Used: **Node.js, TypeScript, Express.js, MongoDB, GitLab CI/CD**
- **Autonomous Parking System** *Hochschule Darmstadt*
    - Developed and calibrated sensors for the RVR vehicle, improving locator precision by **20%** and enhancing IMU accuracy.
    - Designed system architecture and Python-based interfaces to handle real-time data efficiently.
    - Authored comprehensive project documentation to support future development.

Languages

- **Ukrainian:** Native
- **English:** C1 | University of Central Lancashire Language Courses, UK
- **German:** B2.1 | Intensive & Extensive courses, Hochschule Darmstadt
- **Italian:** B1

Achievments

- **The 2023 ICPC Ukraine Western Contest** *Sep 30th, 2023*
- **International Programming Proggy- Buggy Towel Contest** *May 25th, 2023*
- **The 2022 ICPC Ukraine Western Contest** *Oct 22th, 2022*
- **GoITeens Hakathon** | HTML & CSS Website category *Jun 18th – 19th, 2023*
- **Codeforces 1600+ Solved Problems**